

1. Name: MARK ABRAHAM MAGUMBA

2. Proposed Position: ICT, Artificial Intelligence, Data Science and Digital Transformation Consultant

3. Employer: VANGUARD ECONOMICS

4. Date of Birth: 03-02-1985

Nationality: UGANDAN

5. Education

Makerere University, Uganda	PhD in Information Systems	21 st May 2021
University of Sunderland, United Kingdom	Masters in IT Management	December 2009
Makerere University Uganda	Bachelor of Information Technology	27 th September 2007

6. Professional Certification or Membership in Professional Associations:

- AWS Certified Machine Learning – Specialty, AWS Training and Certification, January 2025
- AWS Certified Cloud Practitioner, AWS Training and Certification. August 2024
- Oracle Certified Associate 10g, Oracle Corporation, March 2010
- CIW (Certified international Webmaster), Pearson VUE, June 2009
- CIW professional, Pearson VUE, June 2009

7. Other Relevant Training:

- Applied Data Science Module, Scientific Computing and Python for Data Science, WorldQuant University/ The Data incubator, September 30th, 2019 – 20th March 2020
- Open Science Grid (OSG) User School, University of Wisconsin-Madison (USA), July 2018
- Visiting PhD student at the University of Bergen (Norway), PhD taught courses (INF 283, INF 358, MNF 990), NORHED – HITRAIN project, August 2015 – December 2015
- OpenMRS Power User, Decimal Works, NORHED-HITRAIN project, February 2016
- Microsoft Sharepoint, Microsoft Innovations Centre/ Makerere University College of Computing and Information Sciences – September 2012

8. Countries of Work Experience: Uganda, Republic of Ireland

9. Languages

Language	Proficiency		
	<i>Speaking</i>	<i>Reading</i>	<i>Writing</i>
English	Fluent	Fluent	Fluent
Luganda	Fluent	Fluent	Fair
Lusoga	Native	Fluent	Fair

10. Employment Record

September 2023 To date

Employer: Technological University Dublin, Republic of Ireland

Position Held: Postdoctoral researcher

Summary:

- Conduct research on scalable and energy-efficient AI systems using neural network compression techniques including quantization, pruning, federated learning, collaborative inference, and neural network re-architecting.
- Design and implement machine learning workflows and experimental pipelines for edge AI applications and resource-constrained deployment scenarios.
- Investigate techniques for improving inference efficiency, model scalability, and energy consumption of deep learning systems.
- Collaborate with multidisciplinary research teams and industry stakeholders on AI-focused innovation projects.
- Deliver technical feasibility studies and AI consultancy through industry engagement initiatives.
- Manage Linux-based development infrastructure including software deployment, system administration, and security hardening.
- Publish research findings in peer-reviewed journals and international conferences.

August 2012 – December 2025

Employer: Makerere University College of Computing and Information Sciences, Uganda

Position Held: Lecturer / Assistant Lecturer

Summary:

- Delivered undergraduate and postgraduate courses in Machine Learning, Artificial Intelligence, Data Science, Data Visualization, Applied Statistics, and Research Methods
- Supervised research and software engineering projects involving machine learning systems and analytics.
- Led and participated in funded research projects focused on AI, NLP, disease surveillance, and data analytics.
- Secured and managed research funding from organizations including Google Research and Makerere Research and Innovations Fund.
- Designed practical laboratory exercises and course content covering machine learning, databases, statistics, and scalable computing.

January 2013 – January 2014

Employer: National Forum for People Living with HIV/AIDS Networks in Uganda

Position Held: ICT Consultant

Summary:

- Conducted systems analysis and requirements gathering.
- Designed and implemented IT solutions to improve operational efficiency
- Provided stakeholder-facing technical consultancy and deployment support.

June 2010 – April 2011

Employer: Movit Products Limited

Position Held: ICT Manager

Summary:

- Managed enterprise IT infrastructure, systems administration, security, and software support.
- Coordinated development of ERP and biometric attendance solutions integrated with HR workflows.
- Implemented IT policies, backup strategies, and operational security controls.

11. Detailed Tasks Assigned

Name of Assignment or Project:	Pearson Feasibility Study
Year	2026
Location:	Republic of Ireland
Client:	Pearson Milking Technology/ Enterprise Ireland
Main Project Features:	Feasibility study for proprietary AI powered equipment diagnostics
Position held:	Technical lead
Activities Performed	Solution setup, preliminary experiments, prototyping, model evaluation, patent preparation and final report writing

Name of Assignment or Project:	MyLaw MVP Client Study
Year	2025 – 2026
Location:	Republic of Ireland
Client:	MyLaw.ie/ Enterprise Ireland
Main Project Features:	Feasibility study for proprietary AI powered legal web app.
Position held:	Technical lead
Activities Performed	Requirements analysis, prototyping, prompt engineering, LLM evaluation, security vulnerability assessment and writing final report with recommendations for security implementation, production controls and AI agents strategy

Name of Assignment or Project:	StylAI AI morphing Feasibility Study
Year	2025
Location:	Republic of Ireland
Client:	MyLaw.ie/ Enterprise Ireland
Main Project Features:	Feasibility study for AI body morphing for virtual garment try ons
Position held:	Technical lead
Activities Performed	Evaluation of the state-of-the-Art AI body morphing solutions such as StyleGAN Human, to test feasibility of the client's intended solution. This included generating test data, setting up preprocessing steps like pose extraction with open pose, generating transformed images and contributing to the final report and design of user evaluation study for the client

Name of Assignment or Project:	Oceanaut Marine Drone Design
Year	2024
Location:	Republic of Ireland
Client:	Oceanaut/ Enterprise Ireland

Main Project Features:	Feasibility study for autonomous marine drone
Position held:	Technical lead
Activities Performed	Requirements analysis, hardware design, determination of data and software requirements and contributed to final report with recommendations for hardware and software implementation and data collection and AI model development

Name of Assignment or Project:	BASIS: Broad Avian Species Surveillance with Intelligent Sensing
Year	2023 - 2024
Location:	Uganda
Client:	Google Research Collabs/ Makerere University
Main Project Features:	AI powered, remote passive bio-acoustic monitoring of African bird species in the wild
Position held:	Principle investigator
Activities Performed	Overall project design, coordination, financial and progress reporting. Also responsible for python/PostgreSQL backend development, model evaluation and communicating project results

Name of Assignment or Project:	RAPID: A Real Time Application for Screening Symptoms of Perinatal Depression Among Expectant Mothers
Year	2021 - 2022
Location:	Uganda
Client:	Makerere University Research and Innovations Fund
Main Project Features:	Baseline survey Edinburgh Postnatal Depression Scale (EPDS) survey for Kampala, Uganda and a mobile app implementation of the EPDS
Position held:	Co-PI and Technical Lead
Activities Performed	Technical aspects of the project including design and implementation of the mobile android implementation of the EPDS and data collection app for research assistants, statistical analyses and paper writing

Name of Assignment or Project:	Tracking Critical Care Resources: Development of a web and mobile-based application to improve critical care of COVID-19 patients
Year	2021 - 2022
Location:	Uganda
Client:	Makerere University Research and Innovations Fund
Main Project Features:	A web and mobile-based system for tracking the availability of resources needed in critical management of COVID-19 by facility.
Position held:	Co-PI
Activities Performed	Led database design and requirements analysis

Name of Assignment or Project:	Data-Sanitizer: Semi-Automated Identification and Fact-checking of Viral COVID19 Related Posts Using Online Expert Panels
Year	2020 - 2021
Location:	Uganda

Client:	Makerere University Research and Innovations Fund
Main Project Features:	A web and mobile-based system for automatically detecting and tagging false claims related to COVID from social media feeds
Position held:	PI and technical lead
Activities Performed	Led project design, implementation, coding, solution deployment, evaluation and project reporting

12. List of Key Publications:

- **Magumba, Mark Abraham**, Hasifah Kasujja, and Tonny Justus Oyana. *"Neural Semantic Prediction of Prenatal EPDS subscale 3A Categories Using Socio-Demographic Features."* IEEE Access (2025).
- **Magumba, Mark Abraham**, Steven Davy, and Owais bin Zuber. *"N for Parameter: Efficient Multi-Scale Neural Audio Super Resolution with GAN."* ECAI 2025. IOS Press, 2025. 313-321.
- Namatovu, Hasifah Kasujja, **Mark Abraham Magumba**, and Dickens Akena. "E-Screening for Prenatal Depression in Kampala, Uganda Using the Edinburgh Postnatal Depression Scale: Survey Results." *Online Journal of Public Health Informatics* 17.1 (2025): e51602.
- **Magumba, Mark Abraham**, Tom Denton, and Mutesasira Bashir. *"A bird vocalization dataset of birds in Uganda for automated bio-acoustic monitoring and analysis."* *Data in Brief* 54 (2024): 110433.
- Gichuhi, Haron W., **Magumba, Mak Abraham**, Manish Kumar, and Roy William Mayega. *"A machine learning approach to explore individual risk factors for tuberculosis treatment non-adherence in Mukono district."* PLOS Global Public Health 3, no. 7 (2023): e0001466.
- Namatovu, Hasifah Kasujja, **Magumba, Mak Abraham** and Tonny Justus Oyana. *"A water quality dataset of levels of metal, nutrient and anions in sample water points from sixteen selected urban and rural districts of Uganda."* *Data in Brief* 50 (2023): 109601.
- Namatovu, Hasifah Kasujja, and **Magumba, Mak Abraham**. "Deciphering Barriers and Facilitators of eHealth Adoption in Uganda Using the Systems Thinking Approach-A Systematic Review." *International Conference on e-Infrastructure and e-Services for Developing Countries*. Cham: Springer Nature Switzerland, 2022.
- Namatovu, Hasifah Kasujja, and **Magumba, Mark Abraham**. *"Barriers and Facilitators of eHealth Adoption Among Patients in Uganda—a Quantitative Study."* *International Conference on e-Infrastructure and e-Services for Developing Countries*. Cham: Springer Nature Switzerland, 2022.
- Namatovu, H. K., Semwanga, A. R., Kiberu, V. M., Ndigezza, L., **Magumba, Mark Abraham**, & Kyanda, S. K. (2021, December). *"Barriers and facilitators of eHealth adoption among healthcare providers in Uganda—a quantitative study"*. In *International Conference on e-Infrastructure and e-Services for Developing Countries* (pp. 234-251). Cham: Springer International Publishing.
- **Magumba, Mark Abraham**, and Peter Nabende. *"Evaluation of different machine learning approaches and input text representations for multilingual classification of tweets for disease surveillance in the social web."* *Journal of Big Data* 8, no. 1 (2021): 1-17.

- **Magumba, Mark Abraham**, Peter Nabende, and Ernest Mwebaze. "*Ontology boosted deep learning for disease name extraction from Twitter messages.*" Journal of Big Data 5, no. 1 (2018): 31.
- **Magumba, Mark Abraham**, Peter Nabende, and Ernest Mwebaze. "*Message Classification for Generalized Disease Incidence Detection with Topologically Derived Concept Embeddings.*", in the 2018 International Conference on Data Science, ICDATA'18, CSREA Press 2018
- **Magumba, Mark Abraham**, Peter Nabende, and Ernest Mwebaze. "*Ontology driven machine learning approach for disease name extraction from twitter messages.*" In 2017 2nd IEEE International Conference on Computational Intelligence and Applications (ICCI), pp. 68-73. IEEE, 2017
- **Magumba, Mark Abraham**, and Peter Nabende. "*An Ontology for generalized disease incidence detection on twitter.*" In International Conference on Hybrid Artificial Intelligence Systems, pp: 38-51. Springer, Cham, 2017.

12. Referees

Name	Relationship	Position	Contact
Dr. Peter Nabende	PhD supervisor	Deputy Principal Makerere University College of Computing and Information Sciences	+256772618968 Peter.nabende@gmail.com
Dr. Agnes Nakakawa	Former immediate work supervisor	Chair Department of Information Systems, Makerere University Uganda	+256772690247 agnesnakakawa@gmail.com
Dr. Joseph Balikuddembe	Former work superior	Dean Makerere University School of Computing and Informatics Technology	+256756813535 Jkbaliks@gmail.com
Dr. Steven Davy	Immediate work supervisor	Research Center Manager, Technological University Dublin Center for Sustainable Digital technologies	Steven.davy@tudublin.ie